

vectors and coordinate systems

vectors

- A vector has both magnitude and direction. The magnitude of $\underline{E}$ is a scalar quantity written as $E/|\underline{E}|$. A unit vector $\underline{e}_E$ has the same direction as $\underline{E}$.

$$\underline{E} = E \underline{e}_E$$

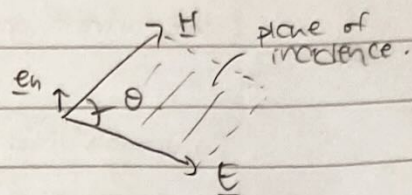
- A vector can be represented in any suitable coord system as a series of component values. eg in Cartesian coords, $\underline{E} = E_x \underline{e}_x + E_y \underline{e}_y + E_z \underline{e}_z$.

- The dot product of $\underline{E}$ and $\underline{H}$ is given by

$$\underline{E} \cdot \underline{H} = |\underline{E}| |\underline{H}| \cos \theta$$

- The cross product of $\underline{E}$ and $\underline{H}$ is given by

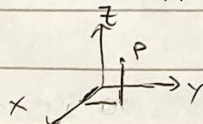
$$\underline{E} \times \underline{H} = |\underline{E}| |\underline{H}| \sin \theta \underline{e}_n$$



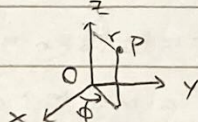
Coordinate systems.

- The key to solving any electromagnetic problem is to use the correct symmetry to simplify the maths. $\rightarrow$ use the correct coordinate system.

- ① Cartesian / rectangular coordinates. (x, y, z)



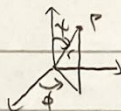
- ② Circular cylindrical coordinates (r, ϕ, z) [wires]



ϕ : azimuthal

$$r = \sqrt{x^2 + y^2}, \quad \phi = \arctan\left(\frac{y}{x}\right), \quad z = z, \quad // \quad x = r \cos \phi, \quad y = r \sin \phi, \quad z = z.$$

- ③ Spherical coordinates (r, ϕ, ψ) [balls, pts, spheres].



ϕ : azimuthal

ψ : zenithal

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \phi = \arctan\left(\frac{y}{x}\right), \quad \psi = \arctan\left(\frac{\sqrt{x^2 + y^2}}{z}\right) // \quad x = r \sin \psi \cos \phi, \quad y = r \sin \psi \sin \phi, \quad z = r \cos \psi$$

Electric force and electric field.

Electric charge.

- The charge of an e^- is $-1.6 \times 10^{-19} \text{ C}$. $\rightarrow$ a unit charge $e = 1.6 \times 10^{-19} \text{ C}$.
- The interaction of charges generates force and force fields.
 - $\rightarrow$ charges of the same sign - repel
 - $\rightarrow$ charges of the opposite sign - attract.

Electrical conductors and insulators.

- In insulators (i.e. plastics, ceramics, dielectrics), charge motion is restricted within the material $\rightarrow$ charge is assumed to be distributed throughout the material (unless told otherwise).
- However, in some cases, there may still be an imbalance of e^- throughout the molecules in the material $\rightarrow$ net overall charge. (dielectrics).
- In conductors, (i.e. metals), charges are free to move throughout the material $\rightarrow$ charge is assumed to be uniformly distributed on the surface of the material (since charges are furthest apart in this configuration).

Coulomb's law

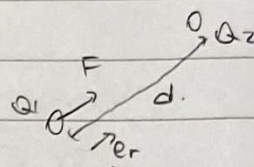
- Coulomb's law states that the electrostatic force F is given by.

$$\underline{F} = k_e \frac{Q_1 Q_2}{r^2} \underline{e}_r = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2} \underline{e}_r$$

where r is the distance between the 2 charges Q_1, Q_2 .

* k_e is Coulomb's constant. $k_e = 8.988 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$.

ϵ_0 is the dielectric permittivity of free space, $\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$.

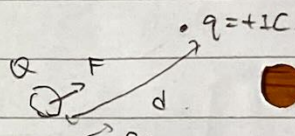


Electric field.

- The electric field $\underline{E}$ represents the lines of force surrounding charges in an electrostatic system. It is the electrostatic force experienced by a unit charge.

$$\underline{E} = k_e \frac{Q}{r^2} \underline{e}_r = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \underline{e}_r$$

and $\underline{F} = q\underline{E}$, where q is a unit test charge.



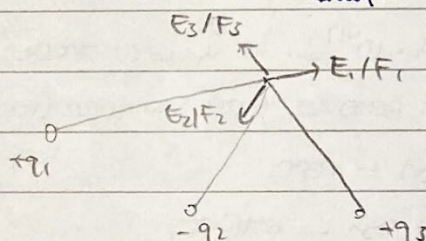
- The density of electric field lines represents the electric field strength, $\underline{E}$.
- Electric field lines always begin and end at a charged particle/pt. on a surface (infinity).

Superposition

- The electrostatic force $\underline{F}$ and electric field $\underline{E}$ are vector quantities $\rightarrow$ the net force and electric field can be found by the vector sum of the contributions of each charge.

$$\underline{F} = \sum \underline{F}_i = \sum \frac{Q_i Q_j}{4\pi\epsilon_0 r_{ij}^2} \underline{e}_{r_{ij}}$$

$$\underline{E} = \sum \underline{E}_i = \sum \frac{Q_i}{4\pi\epsilon_0 r_{ij}^2} \underline{e}_{r_{ij}}$$

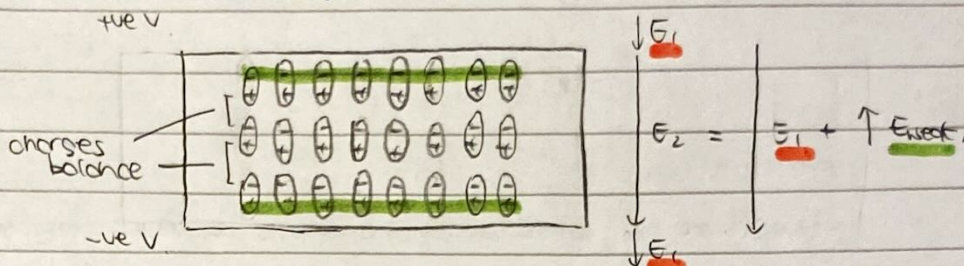


Dielectrics and electric flux density.

Dielectrics

- When charges within a material are subjected to an electric field,
 - ↳ metal: charges are free to move → current flow
 - ↳ dielectric: charges bound inside the molecules but can move within each molecule.
 - charge imbalance (polarisation).

- Consider a dielectric material subject to an electric field E_1 .



- The electric field polarises the molecules within the dielectric, slightly separating the pos and neg charges.
- In the bulk of the dielectric, the pos/neg charges cancel out, but on the surface, a net pos and neg charge appears.
 - ↳ $E_{induced} = -\chi_e E_2$, χ_e is the electrical susceptibility.
- The net charge at the edge sets up weak internal E field $E_{induced}$, which is the opposite dir. of the ext. E field E_1 .

- The net electric field E_2 in the dielectric material is therefore given by

$$(E_2 = E_1 + E_{induced} = E_1 - \chi_e E_2) \quad E_2 = E_1 - E_{induced} \quad \text{so } |E_2| < |E_1|$$

- For a linear material, the E field inside the dielectric material is proportional to the ext. E field E_1 , so

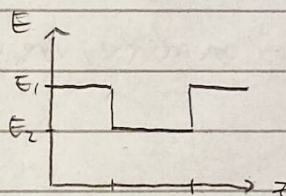
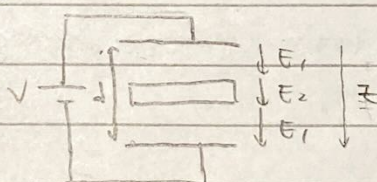
$$\begin{array}{ccc} \text{across air} & & \text{across dielectric} \\ \downarrow & & \downarrow \\ E_1 = E_r E_2 \end{array}$$

where E_r is the rel. permittivity / dielectric constant of the material.

* $E_r \geq 1$. It is assumed that $E_r = 1$ for vacuum/air. [permittivity $\epsilon = \epsilon_0 E_r$].

- This assumes that materials are perfect dielectrics. However in reality, when the E field is large, the e^- will be pulled out of the molecules → material starts conducting. (dielectric breakdown). The dielectric strength E_{max} of a material is defined as the max. E field it can tolerate before breakdown occurs. ($E_{max \text{ air}} = 3 \times 10^6 \text{ Vm}^{-1}$).

- Consider a // plate capacitor w/ a dielectric.



→ E - z graph is discontinuous → define a new quantity: electric flux density D .

Electric flux density

- It is more convenient to use electric flux density $\underline{D}$ rather than E-field $\underline{E}$ to analyse dielectrics.

- The electric flux density $\underline{D}$ is defined as.

$$\underline{D} = \epsilon \underline{E} = \epsilon_0 \epsilon_r \underline{E}$$

$\underline{D}$ is continuous

$\underline{E}$ is directly related to voltage.

- The normal component of electric flux density $\underline{D}$ takes the same value on passing from one dielectric to another $\rightarrow$ flux conservation. (always continuous!)

Gauss law and electric potential.

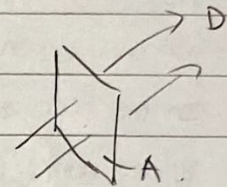
Electric flux.

- Flux is the idea of evaluating the amt. of a vector field that passes through a surface of area A .

- Electric flux Φ_D is given by.

$$\Phi_D = \iint_S \underline{D} \cdot d\underline{A}$$

- If the field is always $\perp$ to the surface, $\Phi_D = DA$.



Gauss's law.

Gaussian surface.

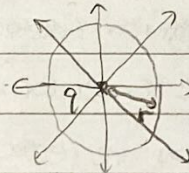
- Gauss's law states that the total electric flux of $\underline{D}$ through any closed surface S equals the charge enclosed by the surface.

$$Q_{enc} = \oiint_S \underline{D} \cdot d\underline{S} = \Phi_D$$

- Gauss's law is very useful because if we know the charge Q , we can calculate $\underline{D}$.

① A point charge (or sphere of charge).

consider $\rightarrow$ pt. charge $+q$



G/L

$$Q_{enc} = \oiint_S \underline{D} \cdot d\underline{S}$$

$$q = D \cdot 4\pi r^2$$

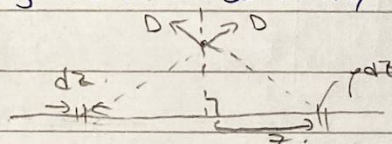
$$D = \frac{q}{4\pi r^2}$$

$\underline{D}$ always $\perp \underline{S}$.

consistent w/ Coulombs law.

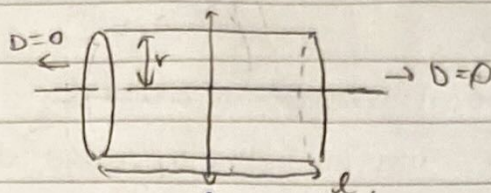
② A line of charge.

Consider an infinitely long wire w/ charge density (charge per unit length) ρ .



Horizontal component of $\underline{D}$ cancels out.

By symmetry, the resultant $\underline{D}$ must act radially outwards at every pt. along the line.



G/L

$$Q_{enc} = \oint_S D \cdot dS$$

$$p \ell = D \cdot (2\pi r \ell)$$

the ends of the cylinder are ignored as flux through them is 0.

$$D = \frac{\rho}{2\pi r}$$

Alternatively, this could be solved using Coulomb's law.

$$R = \sqrt{r^2 + z^2}, \quad \cos\theta = \frac{r}{R} = \frac{r}{\sqrt{r^2 + z^2}}$$

Consider 2 elements either side a distance z away.

$$dq = \rho \delta z$$

Coulomb's law.

$$\frac{\partial V}{\partial z} = \frac{1}{4\pi\epsilon} \frac{\partial}{\partial z} \frac{q}{R^2} \cos\theta \times 2$$

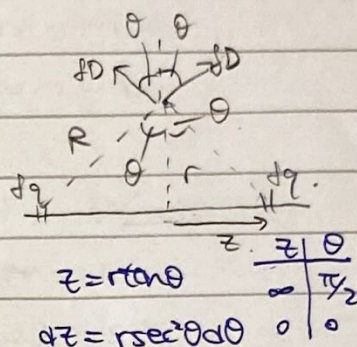
$$\delta D_z = \frac{1}{2\pi} \frac{\rho r}{(r^2 + z^2)^{3/2}} dz$$

$$D = \frac{\rho r}{2\pi} \int_0^\infty \frac{1}{(r^2 + z^2)^{3/2}} dz$$

$$= \frac{\rho r}{2\pi} \int_0^{\pi/2} \frac{1}{(r^2 \sec^2\theta)^{3/2}} r \sec^2\theta d\theta$$

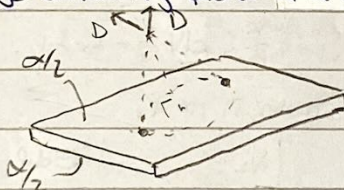
$$= \frac{\rho r}{2\pi} \int_0^{\pi/2} \frac{\cos\theta}{r^2} d\theta$$

$$= \frac{\rho}{2\pi r}$$



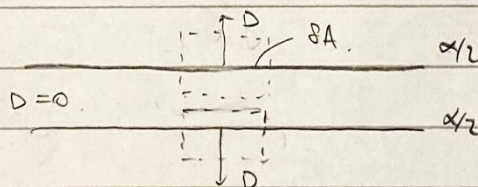
③ Thin uniform metal plate.

Consider an infinitely large conducting plane w/ a charge density (charge per unit area) α .



components of D // to plane cancel out.

By symmetry, the resultant D must be $\perp$ to the plane's surface



By symmetry, $D_{top} = D_{bottom}$. Consider the top surface.

G/L

$$Q_{enc} = \oint_S D \cdot dS$$

$$\frac{\alpha}{2} \delta A = D \delta A$$

$$D = \frac{\alpha}{2}$$

Key results: Point/Sphere: $D = \frac{1}{4\pi\epsilon r^2}$ $E = \frac{1}{4\pi\epsilon r^2}$

Line: $D = \frac{\rho}{2\pi r}$ $E = \frac{\rho}{2\pi\epsilon r}$

Plane: $D = \frac{\alpha}{2}$ $E = \frac{\alpha}{2\epsilon}$

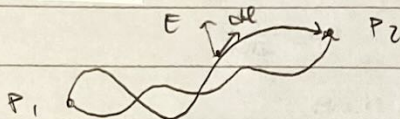
The Faraday cage.

- A simple but important consequence of Gauss's law is the absence of an electric field inside a hollow, charged conducting thinned wall shell.
- When considering a conducting object where charges are free to move, we always assume the charge resides on the surface.
- A Gaussian surface S inside must have 0 flux through it, since it encloses 0 charge.
- Given 0 flux for a finite surface area $\rightarrow$ E -field is 0 everywhere inside the hollow conducting shell.



Potential difference.

- Consider the work req. to move a unit charge from pt. P_1 to P_2 in the presence of an E -field E . At any pt. along the path, we must exert a force $-E$ to hold the charge in eqn.
- The electrostatic potential difference b/w 2 pts. is the work done when a unit free charge is moved from one pt. to another.



same work done for both paths as the force field is conservative.

- Work done in moving distance dl is:

$$\text{work done on the system } dV = - \overset{\text{ext. force}}{E} \cdot dl$$

So total work to move from P_1 to P_2 :

$$V_2 - V_1 = - \int_1^2 E \cdot dl \quad [\text{Eqn. w/ no name}]$$

- Electrostatic potential is defined to be 0 at infinity.

Capacitance

Capacitance.

- Capacitance exists between any pair of conductors which are electrically insulated from each other (Actually everything has a capacitance but it may be extremely small!).
- Capacitance C is the ratio of charge to potential difference applied (units $F(m^2/m^2)$).

$$Q = CV$$

point

$$P = C_L V$$

line

$$X = C_A V$$

plane.

- Strategy to find capacitance:

$\rightarrow$ ① Specify voltages / charges

$\rightarrow$ ② Calculate E (using $D = \epsilon_0 \epsilon_r E$).

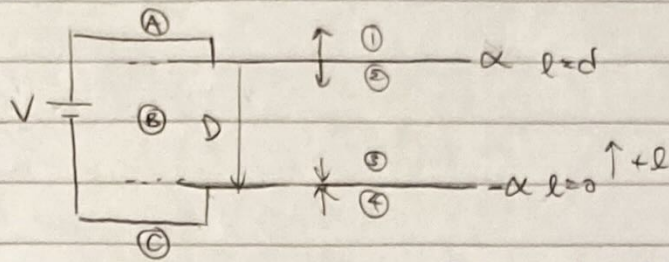
$\rightarrow$ ③ Calculate C (using $C = \frac{Q}{V}$).

use superposition for combinations of pts, lines, planes.

$\rightarrow$ ④ Calculate D (using Gauss' law)

$\rightarrow$ ⑤ Calculate V (using EMU)

Capacitance of a pair of // conducting plates.



At A $\Sigma D = D_1 + D_3$

$$= \frac{\alpha}{2} \uparrow + \frac{\alpha}{2} \downarrow = 0$$

At C $\Sigma D = D_2 + D_4$

$$= \frac{\alpha}{2} \downarrow + \frac{\alpha}{2} \uparrow = 0.$$

At B $\Sigma D = D_2 + D_3$

$$= \frac{\alpha}{2} \downarrow + \frac{\alpha}{2} \downarrow = \alpha \downarrow.$$

$\therefore D=0$ above top plate / below bot. plate ; $D = \alpha \downarrow$ between the plates.

Find E

$$\epsilon_0 E = D$$

$$E = \frac{\alpha}{\epsilon_0}$$

EUN.

$$V_2 - V_1 = - \int_1^2 E \cdot d\ell$$

$$V - 0 = - \int_0^d (-E) \cdot d\ell$$

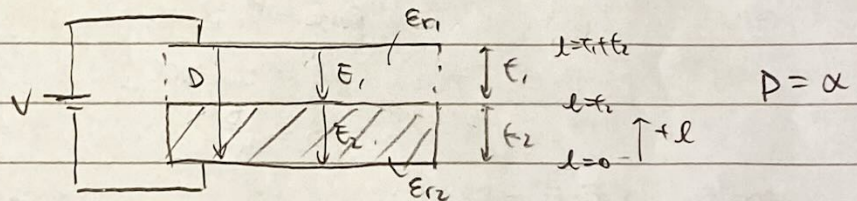
$$V = Ed = \frac{\alpha}{\epsilon_0} d = \frac{Qd}{A\epsilon_0}.$$

Defn of C

$$C = \frac{Q}{V}$$

$$C = \frac{\epsilon_0 A}{d}.$$

Capacitance of a pair of // conducting plates w/ dielectric.



Find E

$$\epsilon_0 \epsilon_1 E_1 = D$$

$$E_1 = \frac{D}{\epsilon_0}$$

$$\epsilon_0 \epsilon_2 E_2 = D$$

$$E_2 = \frac{D}{\epsilon_0 \epsilon_2} = \frac{E_1}{\epsilon_2}$$

EUN

$$V_2 - V_1 = - \int_1^2 E \cdot d\ell$$

$$V - 0 = - \int_0^{t_2} (-E_2) \cdot d\ell - \int_{t_2}^d (-E_1) \cdot d\ell$$

$$V = E_2 t_2 + E_1 t_1 + E_1 t_2 - E_1 t_2$$

$$= \frac{E_1}{\epsilon_2} t_2 + E_1 t_1$$

$$= \frac{D}{\epsilon_0} \left(\frac{t_2}{\epsilon_2} + t_1 \right)$$

$$= \frac{\alpha}{\epsilon_0} \left(\frac{t_2}{\epsilon_2} + t_1 \right) = \frac{Q}{A\epsilon_0} \left(\frac{t_2}{\epsilon_2} + t_1 \right).$$

Defn of C.

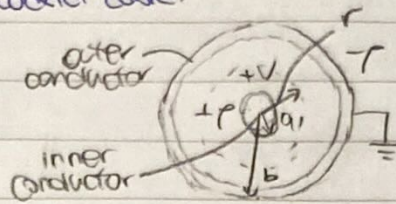
$$C = \frac{Q}{V}$$

$$C = \frac{\epsilon_0 A}{t_1 + \frac{t_2}{\epsilon_2}} = \frac{\epsilon_0 \epsilon_2 A}{\epsilon_2 t_1 + t_2}$$

check $t_1=0$, $C = \frac{\epsilon_0 \epsilon_2 A}{t_2}$

$t_2=0$, $C = \frac{\epsilon_0 A}{t_1}$

capacitance of air-filled coaxial cable.



G/L

$$Q_{enc} = \oint_S \mathbf{D} \cdot d\mathbf{S}$$

$$\rho l = D \cdot (2\pi r l)$$

$$D = \frac{\rho}{2\pi r}$$

Find E

$$\epsilon_0 E = D$$

$$E = \frac{\rho}{2\pi \epsilon_0 r}$$

ENN

$$V_2 - V_1 = - \int_1^2 \mathbf{E} \cdot d\mathbf{l}$$

$$0 - V = - \int_a^b \frac{\rho}{2\pi \epsilon_0 r} dr$$

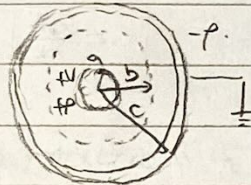
$$V = \frac{\rho}{2\pi \epsilon_0} \ln\left(\frac{b}{a}\right)$$

Defn of C

$$C_l = \frac{Q}{V} = \frac{\rho l}{V}$$

$$C_l = \frac{2\pi \epsilon_0}{\ln\left(\frac{b}{a}\right)}$$

capacitance of coaxial cable w/ two dielectrics



G/L

$$Q_{enc} = \oint_S \mathbf{D} \cdot d\mathbf{S}$$

$$\rho l = D (2\pi r l)$$

$$D = \frac{\rho}{2\pi r}$$

Find E

$$\epsilon_0 \epsilon_1 E_1 = D$$

$$E_1 = \frac{D}{2\pi \epsilon_0 \epsilon_1 r}$$

$$\epsilon_0 \epsilon_2 E_2 = D$$

$$E_2 = \frac{D}{2\pi \epsilon_0 \epsilon_2 r}$$

ENN

$$V_2 - V_1 = - \int_1^2 \mathbf{E} \cdot d\mathbf{l}$$

$$0 - V = - \left[\int_a^b \frac{D}{2\pi \epsilon_0 \epsilon_1 r} dr + \int_b^c \frac{D}{2\pi \epsilon_0 \epsilon_2 r} dr \right]$$

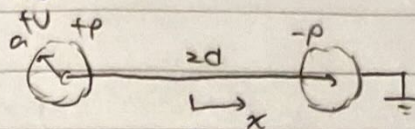
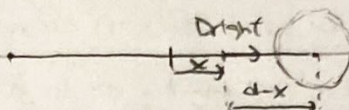
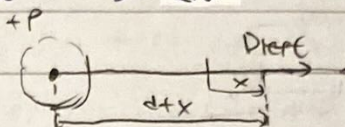
$$V = \frac{\rho}{2\pi \epsilon_0} \left[\frac{1}{\epsilon_1} \ln\left(\frac{b}{a}\right) + \frac{1}{\epsilon_2} \ln\left(\frac{c}{b}\right) \right]$$

Defn of C

$$C_l = \frac{Q}{V} = \frac{\rho l}{V}$$

$$C_l = \frac{2\pi \epsilon_0}{\frac{1}{\epsilon_1} \ln\left(\frac{b}{a}\right) + \frac{1}{\epsilon_2} \ln\left(\frac{c}{b}\right)}$$

capacitance of two parallel wires

D due to an infinitely long wire : $D = \frac{p}{2\pi r}$ 

$$\Sigma D = D_{left} + D_{right}$$

$$= \frac{p}{2\pi(d+x)} + \frac{p}{2\pi(d-x)}$$

Find E

$$\epsilon_0 E = D$$

$$E = \frac{p}{2\pi\epsilon_0(d+x)} + \frac{p}{2\pi\epsilon_0(d-x)}$$

EIN

$$V_2 - V_1 = - \int_1^2 E dl$$

$$0 - V = - \int_a^{2d-a} \frac{p}{2\pi\epsilon_0} \left[\frac{1}{d+x} + \frac{1}{d-x} \right] dx$$

$$V = \frac{p}{\pi\epsilon_0} \ln\left(\frac{2d-a}{a}\right)$$

Defn of C

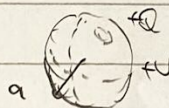
$$C = \frac{Q}{V} = \frac{pL}{V}$$

$$= \frac{\pi\epsilon_0}{\ln\left(\frac{2d-a}{a}\right)}$$

since $a \ll d$, $2d-a \approx 2d$

$$\therefore C \approx \frac{\pi\epsilon_0}{\ln\left(\frac{2d}{a}\right)}$$

capacitance of a sphere.



GIL

$$Q_{enc} = \oint D \cdot dS$$

$$Q = D \cdot (4\pi r^2)$$

$$D = \frac{Q}{4\pi r^2}$$

Find E

$$\epsilon_0 E = D$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

EIN

$$V_2 - V_1 = - \int_1^2 E dr$$

$$0 - V = - \int_a^\infty \frac{Q}{4\pi\epsilon_0 r^2} dr$$

$$V = \frac{Q}{4\pi\epsilon_0 a}$$

Defn of C

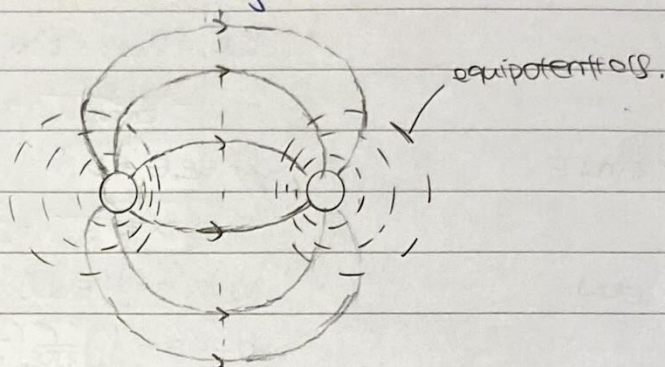
$$C = \frac{Q}{V}$$

$$C = 4\pi\epsilon_0 a$$

Equipotentials

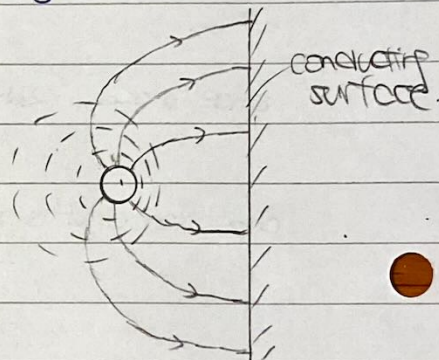
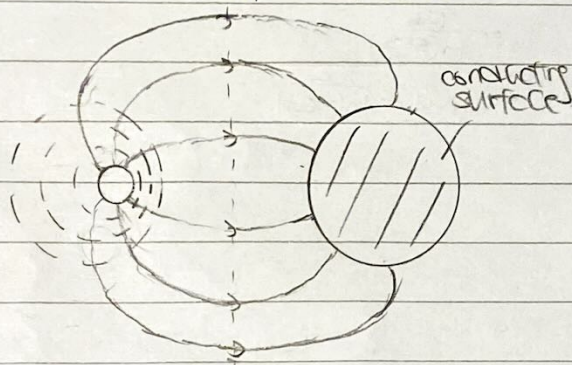
Equipotentials.

- An equipotential is a line (surface) of constant potential.
- We can move a pt. charge along an equipotential w/o doing work against electric fields.
- Equipotential lines are $\perp$ to electric field lines - if a charge travels $\perp$ to an electric field, no work is done by / against the field.
- Consider two equal and opposite charges $+Q$ and $-Q$.
The equipotentials are circles of increasing radius.



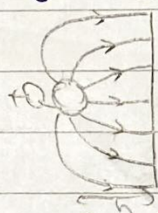
The equipotential midway between the conductors is a line (circle w/ infinite radius).

- We can replace any equipotential w/ a conducting surface.

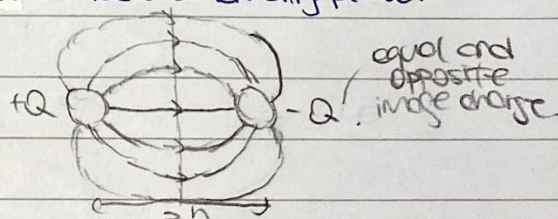


Method of images.

- We can replace a conducting surface w/ an image charge. (that gives same equipotentials)
- Consider a charge $+Q$ at a height h next to a metal conducting plate.



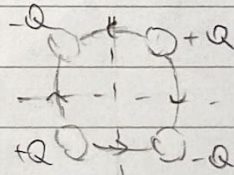
→



- Consider a charge $+Q$ $\perp$ to a right angle conductor.



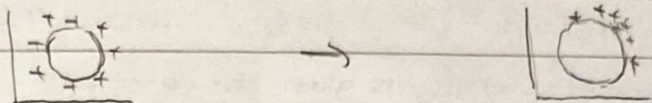
→



For Personal Use Only -bkwk2

- This method is only an approximation as we assume the charged sphere to behave as if it were uniformly charged.

- In practice, the metal plane will distort the charge distribution of the sphere.



- An exact sol'n requires finite element and finite difference methods.

- e.g.: Voltage of sphere above a conducting plane.

A metal sphere of radius $a = 5\text{mm}$ is centred at a height $h = 10\text{mm}$ from an earthed conducting sheet. The electrical breakdown strength of air (E_{max}) is $3 \times 10^6 \text{Vm}^{-1}$.

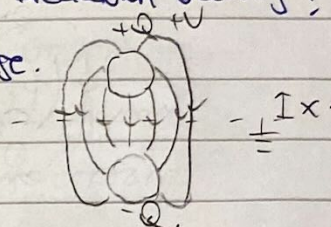
What is the max voltage / charge applied to the sphere w/o breakdown occurring?

First, we replace the conducting sheet w/ an image charge.

For a sphere, $E = \frac{Q}{4\pi\epsilon_0 r^2}$, so by superposition,

$$E = -\left[\frac{Q}{4\pi\epsilon_0 (h-x)^2} + \frac{Q}{4\pi\epsilon_0 (h+x)^2} \right]$$

$\uparrow$ $\uparrow$ $\uparrow$
 E field dir. is opposite to x dir. due to real charge at $x=h$ due to image charge at $x=-h$



As E field is max at the edge of a charged sphere, sub $x = h-a$.

$$|E_{\text{max}}| = \frac{Q}{4\pi\epsilon_0 a^2} + \frac{Q}{4\pi\epsilon_0 (2h-a)^2}$$

Substituting $a = 5\text{mm}$, $h = 10\text{mm}$, $|E_{\text{max}}| = 3 \times 10^6 \text{Vm}^{-1}$, $Q = 7.5\text{nC}$.

$$V_2 - V_1 = -\int_1^2 E \cdot dl$$

$$V - 0 = -\int_0^{h-a} \left[\frac{Q}{4\pi\epsilon_0 (h-x)^2} + \frac{Q}{4\pi\epsilon_0 (h+x)^2} \right] dx$$

$$V = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{2h-a} \right)$$

Substituting $a = 5\text{mm}$, $h = 10\text{mm}$, $Q = 7.5\text{nC}$, $V = 7000\text{V}$.

* Note, w/o the metal plane, $E = \frac{Q}{4\pi\epsilon_0 a^2}$ so $|E_{\text{max}}| = \frac{Q}{4\pi\epsilon_0 a^2}$ and $V = \frac{Q}{4\pi\epsilon_0 a}$.

- substituting $a = 5\text{mm}$, $|E_{\text{max}}| = 3 \times 10^6 \text{Vm}^{-1}$, $Q = 8.34\text{nC}$, $V = 15000\text{V}$.

i.e. as we move a conductor closer to a charged sphere, the voltage req. for breakdown drops.

Electrostatic energy and forces

Energy stored in an electric field.

- Consider the work done δW in adding a charge δq to a capacitor at voltage V .

$$I = \frac{\delta q}{\delta t}, \quad V = \frac{W}{q}, \quad P = IV \quad // \quad \delta W = P \delta t = \frac{\delta q}{\delta t} \frac{W}{q} \delta t = \frac{1}{2} q \delta q$$

$$\therefore W = \int_0^Q \frac{1}{2} q \, dq = \frac{1}{2} Q^2$$

As $Q = CV$, $W = \frac{Q^2}{2C} = \frac{1}{2} QV = \frac{1}{2} CV^2$. $\rightarrow$ E field can store energy.

* The expression above applies to any capacitor of any geometry.

For Personal Use Only -bkwk2

Energy stored in a // plate capacitor.

- Consider a // plate capacitor w/ area A and plate separation x . We neglect any fringe effects at the edges of the plate.

$$V = Ex, \quad C = \frac{\epsilon_0 \epsilon_r A}{x}, \quad W = \frac{1}{2} CV^2 \rightarrow W = \frac{1}{2} \frac{\epsilon_0 \epsilon_r A}{x} E^2 x^2 = \frac{1}{2} \epsilon_0 \epsilon_r E^2 (Ax).$$

$\therefore$ The energy density stored in the electric field is given by $\frac{W}{Ax} = \frac{1}{2} \epsilon_0 \epsilon_r E^2$

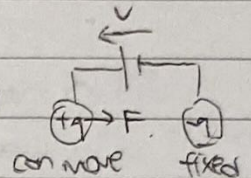
Electrostatic force (Virtual work method).

- consider a system consisting of two arbitrary charged conductors:

$\rightarrow$ one of the conductors is fixed (cannot move)

$\rightarrow$ The conductors have a charge of $+q$ and $-q \rightarrow$ they experience a Coulomb force F

$\rightarrow$ The battery between the conductors keep the potential V constant

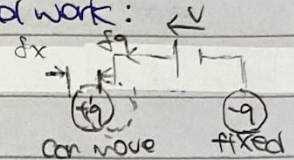


- To find the Coulomb force F , we use the method of virtual work:

$\rightarrow$ Apply a virtual force F^* equal and opposite to F so

the conductors are held in static eqm.

$\rightarrow$ Imagine the moveable conductor moves a small distance δx towards the other.



- consider the energy changes,

$\rightarrow$ Mechanical work is done on the system $-F^* \delta x$.

$\rightarrow$ The battery does work on the system $V \delta q$ by providing an extra charge δq (so V is const)

$\rightarrow$ There is a net change in electrostatic energy of $\frac{1}{2} V \delta q$. ($W = \frac{1}{2} QV$, so $\delta W = \frac{1}{2} V \delta q$).

- Using conv. of mechanical energy, (considering the system).

$$-F \delta x + V \delta q = -\frac{1}{2} V \delta q$$

$$F \delta x = \frac{1}{2} V \delta q$$

$$F = \frac{1}{2} V \frac{\delta q}{\delta x}$$

As $q = CV$, $\frac{\delta q}{\delta x} = V \frac{\partial C}{\partial x}$, so $F = \frac{1}{2} V^2 \frac{\partial C}{\partial x}$

useful as we just need to know V and $\frac{\partial C}{\partial x}$.
(no need to know q, r)
* To find $\frac{\partial C}{\partial x}$, we need $C = f(x)$.

Force between capacitor plates.

- For a // plate capacitor, $C = \frac{\epsilon_0 \epsilon_r A}{x}$, so $\frac{\partial C}{\partial x} = -\frac{\epsilon_0 \epsilon_r A}{x^2}$.

$$F = \frac{1}{2} V^2 \frac{\partial C}{\partial x}$$

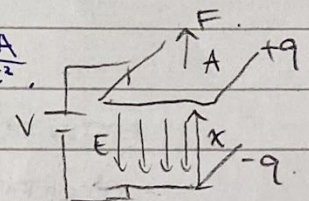
$$= \frac{1}{2} V^2 \left(-\frac{\epsilon_0 \epsilon_r A}{x^2} \right)$$

$$= -\frac{1}{2} V^2 \frac{C}{x}$$

$$F = -\frac{1}{2} qE$$

$$(C = \frac{\epsilon_0 \epsilon_r A}{x})$$

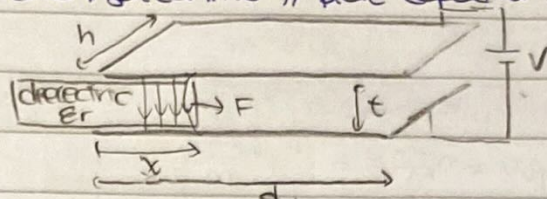
$$(q = CV, E = \frac{V}{x})$$



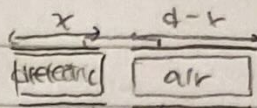
* Note the electrostatic force F is not qE .

* The -ve sign means the force is attractive (opposite the dir. of the x).

Force on dielectric slab inserted into // plate capacitor.



- The system is essentially two capacitors connected in //



$$\therefore C_{eq} = \frac{\epsilon_0 \epsilon_r x h}{t} + \frac{\epsilon_0 (d-x) h}{t} \rightarrow \frac{\partial C}{\partial x} = \frac{\epsilon_0 \epsilon_r h}{t} - \frac{\epsilon_0 h}{t} = \frac{\epsilon_0 h}{t} (\epsilon_r - 1)$$

$$F = \frac{1}{2} V^2 \frac{\partial C}{\partial x}$$

$$F = \frac{1}{2} V^2 \frac{\epsilon_0 h}{t} (\epsilon_r - 1)$$

Electric current

Current.

- Any conductor has delocalised e^- that can move freely about the conductor.
- If an ext. E field is applied to the conductor, the delocalised e^- will accelerate under the applied force and drift in a single direction, creating a current.
- Current I is defined as the rate of change of charge through a fixed area wrt. time, i.e. $I = \frac{dq}{dt}$

Current density.

- current density J is defined as.

$$dI = J \cdot dS, \text{ or } I = \int_S J \cdot dS \quad (J = \frac{dI}{dS})$$

- If the conductor has a uniform cross section, $J = \frac{I}{S}$
- Considering a conducting volume w/ charge density ρ_v that moves at a drift velocity u , then the current density J is given by

$$J = \rho_v u$$

Resistance.

- Considering conduction currents, e^- experience a force F when an E field E is applied, $F = -eE$.
- This force moves the e^- along the wire but the many collisions slow them down to an average drift velocity u . If the e^- have mass m and the average time between collisions is τ , the force is given by

$$F = \frac{mu}{\tau} = -eE$$

$$\therefore u = -\frac{e\tau}{m} E$$

For Personal Use Only -bkwk2

- If there are $n e^-$ in a unit volume, then charge density ρ_v is simply

$$\rho_v = -ne.$$

- Using the defn of current density $\underline{J}$,

$$\underline{J} = \rho_v \underline{u} = \frac{ne^2}{m} \underline{E} = \sigma \underline{E} \quad [\text{row form of drmk law}].$$

where σ is the conductivity of the conductor.

- For a non-perfect conductor, we can define the imperfections in the conductor as resistance R , where $R = \frac{V}{I}$.

- For a non-perfect conductor w/ uniform cross-sectional area A and length l ,

$$R = \frac{V}{I} = \frac{V}{JA} = \frac{El}{\sigma EA} = \frac{l}{\sigma A}$$

- We can define resistivity ρ as $\rho = \frac{1}{\sigma}$, so

$$R = \rho \frac{l}{A}$$

For Personal Use Only -bkwk2

Magnetic flux density

- Magnetic flux density is a vector denoted as $\underline{B}$
- Magnetic flux density $\underline{B}$ has units T or Wb m^{-2}

- Biot-Savart Law states that a short element $d\vec{l}$ of wire carrying a current I produces a magnetic flux density $d\vec{B}$ at a position described by the vector $\vec{r}$ is

- In scalar form,

Use R4 rule to determine the dir. of B

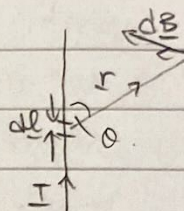
$$dB = \frac{\mu_0 I \sin \theta \, dl}{4\pi r^2}$$

where θ is the angle between the vectors $\underline{r}$ and $\underline{dl}$

* μ_0 is the permeability of free space, $\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$

For a moving charge,

$$\underline{B} = \frac{\mu_0}{4\pi} \frac{q \underline{v} \times \underline{r}}{r^3}$$



Amperean loop.

- Ampère's Law states that for any closed loop, $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$ (just for air/free space)

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$$

$$(I_{enc} = NI)$$

where $\oint \mathbf{B} \cdot d\mathbf{l}$ is the circulation of $\mathbf{B}$ and I_{en} is the current enclosed in the closed loop.

- Alternatively, vary magnetic field intensity H (general, inc. nonlinear materials)

$$\oint_C \underline{H} \cdot d\underline{u} = NI$$

where $\oint \vec{H} \cdot d\vec{l}$ is the circulation of $\vec{H}$ and NI is the magnetomotive force (MMF).

Magnetic flux density around a straight wire.

⑦ Brot-Säure Law.

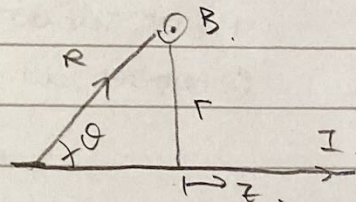
$$dB = \frac{\mu_0 I \sin \theta dz}{4\pi r^2}$$

$$B = \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} \frac{r \sqrt{1+z^2}}{r^4 z^2} dz$$

$$= \frac{\mu_0 I}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{r'}{(r^2 \sec^2 \alpha)^{3/2}} (r \sec^2 \alpha d\alpha)$$

$$= \frac{\mu_0 I}{4\pi c} \int_{-\pi/2}^{\pi/2} \frac{\cos \alpha}{r} d\alpha.$$

$$B = \frac{\mu_0 I}{2\pi r}$$



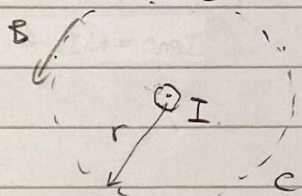
$$R^2 = z^2 + r^2, \quad z = r \tan \alpha$$

(ii) Ampère's Law.

$$\oint \underline{B} \cdot d\underline{l} = \mu_0 I_{enc}$$

$$B(\Sigma T_r) = M.I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

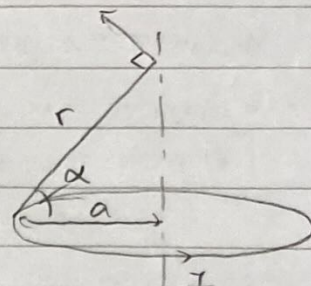


Magnetic flux density on axis of a wire loop.

① Biot-Savart Law

Note B is axial only (horizontal components cancel)

$$\begin{aligned} dB &= \frac{\mu_0 I \sin \theta dl}{4\pi r^2} \quad \text{axial B only} \\ B &= \frac{\mu_0 I \sin \theta}{4\pi r^2} \int dl \cdot \cos \theta \\ &= \frac{\mu_0 I}{4\pi(a^2+z^2)} (2\pi a) \left(\frac{a}{a^2+z^2} \right) \\ B &= \frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}} \end{aligned}$$

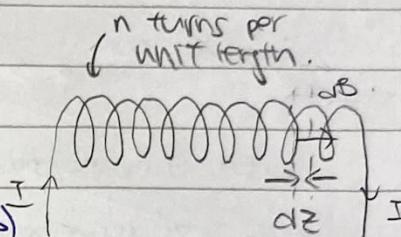


Magnetic flux density inside a solenoid.

① Biot-Savart Law

Consider the magnetic flux density dB produced by the short section dz .

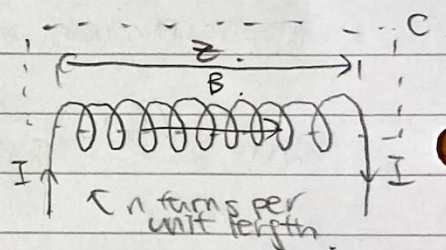
using superposition (valid for infinitesimally small sections)



$$\begin{aligned} dB &= \frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}} \cdot n dz \quad (\text{derived using Biot-Savart Law}) \\ B &= \frac{\mu_0 I a^2 n}{2} \int_{-a}^{a} \frac{1}{(a^2+z^2)^{3/2}} dz \quad z = a \tan \alpha \\ &= \frac{\mu_0 I a^2 n}{2} \int_{-\pi/2}^{\pi/2} \frac{1}{(a \sec^2 \alpha)^{3/2}} (a \sec^2 \alpha d\alpha) \quad dz = a \sec^2 \alpha \\ &= \frac{\mu_0 I n}{2} \int_{-\pi/2}^{\pi/2} \cos \alpha d\alpha \\ B &= \mu_0 n I \end{aligned}$$

② Ampère's Law

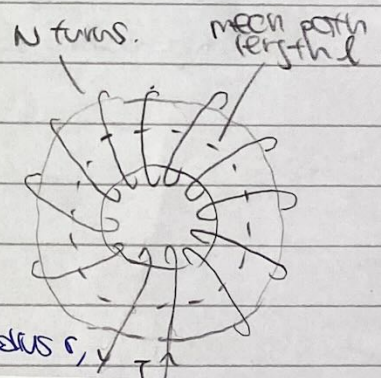
$$\begin{aligned} \oint_C \underline{B} \cdot d\underline{l} &= \mu_0 I_{enc} \\ B \oint &= \mu_0 (n \oint) I \\ B &= \mu_0 n I \end{aligned}$$



Magnetic flux density inside a toroidal coil.

① Ampère's Law

$$\begin{aligned} \oint \underline{B} \cdot d\underline{l} &= \mu_0 I_{enc} \\ B(2\pi r) &= \mu_0 NI \\ B &= \frac{\mu_0 NI}{2\pi r} \end{aligned}$$



* If we consider the mean path length l , at a certain radius r ,

$$2\pi r = l \quad \text{so} \quad B = \frac{\mu_0 NI}{l} = \mu_0 n I$$

Magnetic force

- The magnetic force $\underline{F}$ due to a charge q moving at velocity $\underline{v}$ in magnetic field is

$$\underline{F} = q(\underline{v} \times \underline{B})$$

- The magnetic force $\underline{F}$ of a wire w/ length $\underline{l}$ carrying current I in magnetic field is

$$\underline{F} = I(\underline{l} \times \underline{B})$$

* Note $I = \frac{q}{t} Av = \frac{q}{A \Delta t} Av = \frac{qv}{\Delta t}$. $\rightarrow$ both eqns are equivalent.

- The magnetic force $\underline{F}$ between two // current carrying wires is given by,

$$F = B_2 I_1 l$$

$$= \frac{\mu_0 I_2}{2\pi r} I_1 l$$

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$$

Magnetic flux and self inductance.

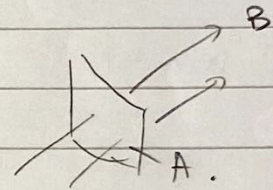
Magnetic flux

- Flux is the idea of evaluating the amt. of a vector field that passes through a surface of area A .

- Magnetic flux Φ_B is given by

$$\Phi_B = \iint_S \underline{B} \cdot d\underline{A}$$

- If the field is always $\perp$ to the surface, $\Phi_B = BA$.



Flux linkage.

- In solenoids and inductors, we have many turns and each turn is effectively another area, so we sum the flux for each turn to get the flux linkage Φ'_B .

$$\Phi'_B = N\Phi_B$$

- For a uniform flux density B passing through N turns each of area A , the flux linkage Φ'_B is given by $\Phi'_B = N\Phi_B = NBA$.

Faraday's law of electromagnetic induction.

- Faraday's Law of electromagnetic induction states that the emf induced $\mathcal{E}$ is prop. to the rate of change of flux linking the circuit.

$$\mathcal{E} = \frac{d\Phi'}{dt}$$

- Lenz's law states that the emf induced by the change in flux is in the opposite dir.

$$\mathcal{E} = -\frac{d\Phi'}{dt}$$

(i.e. the induced emf opposes the flux that generated it).

* Practically, we ignore the - sign and figure out the dir. of $\mathcal{E}$ in the end.

For Personal Use Only -bkwk2

self inductance.

- self inductance L is the ratio of flux linkage to the current generating the flux. (units: H).

$$L = \frac{\Phi_B}{I}$$

- strategy to find self-inductance:

use superposition for combinations of wires, loops.

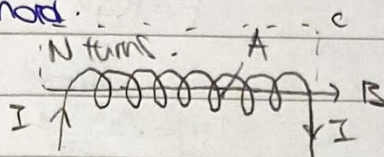
↳ ① specify currents

↳ ② calculate B (using Ampère's Law)

↳ ③ calculate Φ'_B . (using $\Phi'_B = N \int B \cdot dA$)

↳ ④ Calculate L (using $L = \frac{\Phi'_B}{I}$).

self inductance of a solenoid.



A/L

$$\oint_C B \cdot dl = \mu_0 I_{enc}$$

$$B l = \mu_0 N I$$

$$B = \mu_0 \frac{N}{l} I$$

Flux Φ'

$$\Phi'_B = N B A$$

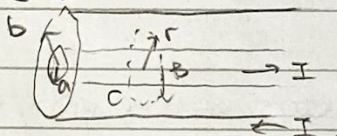
$$= \mu_0 \frac{N^2}{l} I A$$

Defn of L

$$L = \frac{\Phi'_B}{I}$$

$$L = \frac{\mu_0 N^2 A}{l}$$

self inductance of a coaxial cable.



A/L

$$\oint_C B \cdot dl = \mu_0 I_{enc}$$

$$B(2\pi r) = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

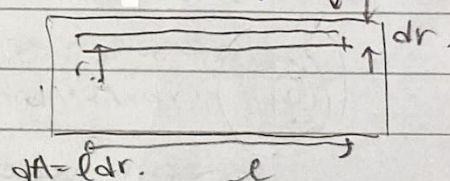
Flux Φ'

$$\Phi'_B = \int_a^b B \cdot dA$$

$$= \int_a^b \frac{\mu_0 I}{2\pi r} l dr$$

$$= \frac{\mu_0 I l}{2\pi} \ln\left(\frac{b}{a}\right)$$

side view
(for $a < r < b$).



Defn of L

$$L = \frac{\Phi'_B}{I}$$

$$L = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right)$$

* Note self inductance per unit length L_ℓ is given by $L_\ell = \frac{L}{l}$.

so for a coaxial cable, $L_\ell = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$.

recall that for a coaxial cable, $C_\ell = \frac{2\pi\epsilon_0}{\ln\left(\frac{b}{a}\right)}$.

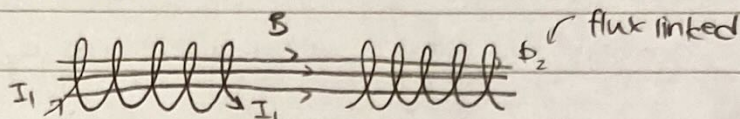
so $\sqrt{\frac{L_\ell}{C_\ell}} \propto \sqrt{\frac{\mu_0}{\epsilon_0}} = Z_0 = \frac{E}{H}$, where Z_0 is the characteristic impedance of free space.

and $\frac{1}{\sqrt{L_\ell C_\ell}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$, where c is the speed of light in free space.

Mutual inductance.

Mutual inductance.

- When two electrical circuits are close to one another, a current I_1 in one may cause magnetic flux ϕ_2 to link the other.

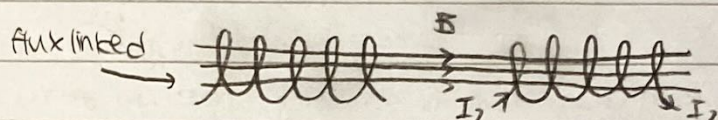


- Assuming linear behaviour, flux ϕ_2 must be prop. to current I_1 ,

$$\phi_2 = M_{12} I_1$$

where the const. of prop. M_{12} is the mutual inductance between the circuits.

- Similarly, w/ the roles of the above circuits reversed, a current I_2 may cause magnetic flux ϕ_1 to link the other.



$$\phi_1 = M_{21} I_2$$

- It can be shown that the const. of prop. M_{12} and M_{21} are equal.

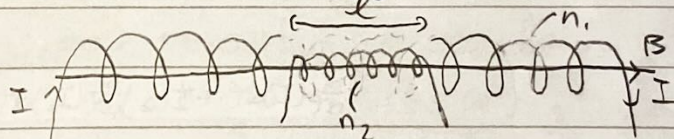
$$M_{12} = M_{21} = M$$

The quantity M is mutual inductance (units: H).

Mutual inductance of concentric solenoids

- Consider a pair of concentric coils. the outer coil has n_1 turns per unit length; the inner coil has n_2 turns per unit length, CSA A and length l .

* Assume diameter of inner coil is small enough that B is const. over its CSA.



Magnetic flux density B due to outer coil $B = \mu_0 n_1 I$

Magnetic flux ϕ_2 through 1 turn of inner coil $\phi_2 = BA = \mu_0 n_1 I A$.

Magnetic flux linked ϕ_2' through $n_2 l$ turns of inner coil $\phi_2' = n_2 l \phi_2 = \mu_0 n_1 n_2 l I A$.

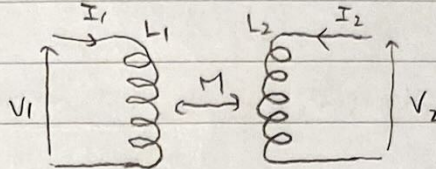
Mutual inductance $M = \frac{\phi_2'}{I} = \mu_0 n_1 n_2 l A$.

(similar strategy to finding the self-inductance but we want to find ϕ_2' rather than ϕ_1' for a current I_1).

For Personal Use Only -bkwk2

Magnetic energy in current-carrying circuits

- When currents I are established in an electrical circuit (change in I), there is a corresponding change in magnetic flux (density) B/d , hence an induced emf ϵ . This means the voltage source must do work against this emf $\rightarrow$ energy stored in circuit.
- Consider two circuits w/ self inductances L_1, L_2 and mutual inductance M .



- The flux linking each circuit is a combination of the flux generated by its own current and the flux generated by the other current.

$$\begin{aligned}\Phi_1 &= L_1 I_1 + M I_2 \\ \Phi_2 &= L_2 I_2 + M I_1\end{aligned}$$

self mutual

- As the flux in each circuit is changing, there will be induced emfs V_1 and V_2 in each circuit, w/ the magnitude given by

$$\begin{aligned}V_1 &= \frac{d\Phi_1}{dt} = \frac{d}{dt}(L_1 I_1 + M I_2) = L_1 \frac{dI_1}{dt} + M \frac{dI_2}{dt} \\ V_2 &= \frac{d\Phi_2}{dt} = \frac{d}{dt}(L_2 I_2 + M I_1) = L_2 \frac{dI_2}{dt} + M \frac{dI_1}{dt}\end{aligned}$$

- The rate of change electrical energy $\frac{dW}{dt}$ is given by

$$\begin{aligned}\frac{dW}{dt} &= P = I_1 V_1 + I_2 V_2 \\ &= I_1 (L_1 \frac{dI_1}{dt} + M \frac{dI_2}{dt}) + I_2 (L_2 \frac{dI_2}{dt} + M \frac{dI_1}{dt}) \\ &= L_1 I_1 \frac{dI_1}{dt} + L_2 I_2 \frac{dI_2}{dt} + M (I_1 \frac{dI_2}{dt} + I_2 \frac{dI_1}{dt}) \\ &= \frac{1}{2} L_1 \frac{d}{dt}(I_1^2) + \frac{1}{2} L_2 \frac{d}{dt}(I_2^2) + M \frac{d}{dt}(I_1 I_2)\end{aligned}$$

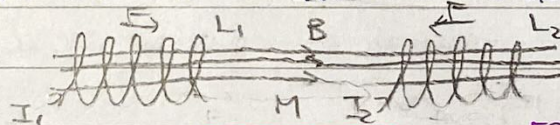
- Integrating wrt time gives the total energy stored

$$W = \frac{1}{2} L_1 \int d(I_1^2) + \frac{1}{2} L_2 \int d(I_2^2) + M \int d(I_1 I_2)$$

$$W = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2$$

i.e. the total energy in the circuit is the sum of the individual inductances ($\frac{1}{2} L_1 I_1^2$) plus the energy of interaction b/w the circuits from the mutual inductance ($M I_1 I_2$).

- The method of virtual work can be applied to this expression to find the force F .



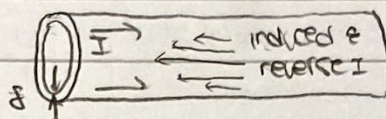
$$F = I_1 I_2 \frac{\partial M}{\partial x}$$

Force F symmetric wrt. interchanging currents I_1, I_2

* In practice, this is a difficult expression to use as it is usually to define the mutual inductance $M(x)$.

The skin effect

- Consider a cylindrical conducting wire along which a high frequency AC current ($f > 100\text{kHz}$)
- Inside the wire, there will be circular lines of magnetic flux, and these will also be oscillating at high frequency
- As a result of Faraday's law, an emf will be induced along the wire: As a result of Lenz's law, this emf opposes the original change in flux, setting up a reverse current.
- At the centre of the wire, magnetic flux Φ is max, therefore a larger reverse current is produced $\rightarrow$ current tends to flow near the surface of the wire.



- In this course, we consider that all currents pass through the thin layer of thickness δ (skin depth) near the surface of the wire. *Actually, at skin depth δ , the current drops to $1/e$ of the max. current.*
- The higher the frequency, the larger the $\frac{d\Phi}{dt}$, so the larger the emf and cancellation of currents, thus the smaller the skin depth.
- The skin depth δ is given by the expression

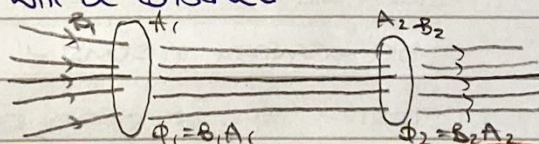
$$\delta = \sqrt{\frac{1}{\pi \mu_0 \sigma f}}$$

- A smaller skin depth δ means a higher resistance in the wire R since the current has a smaller cross section A through which to flow ($A = \pi[r^2 - (r-\delta)^2]$, $R = \rho \frac{l}{A}$)

Magnetic materials

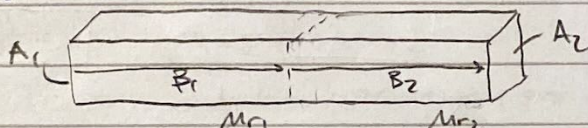
Conservation of magnetic flux

- Lines of magnetic flux density B always form closed loops. i.e. $\nabla \cdot B = 0$
(In contrast to E fields, which always begin/end on charges $\rightarrow$ no magnetic charges (monopoles))
- If we have a region of space where the B lines are contained (confined), then the flux of B across the area will be conserved



$$\iint_{A_1} B \cdot dS = \Phi_1 = \Phi_2 = \iint_{A_2} B \cdot dS$$

- This means given that flux travels from a magnetic material into another magnetic material w/o significant divergence, then flux will be conserved.

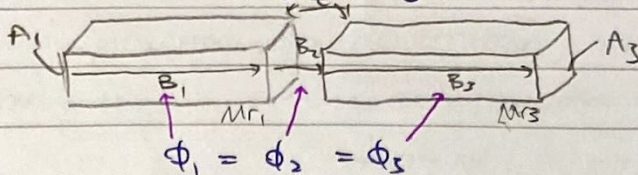


$$\text{If } A_1 = A_2 = A, \quad \Phi_1 = \Phi_2 \rightarrow B_1 A = B_2 A \rightarrow B_1 = B_2$$

i.e. cons. of B as long as it passes through the same CSA A .

For Personal Use Only -bkwk2

- Consider two magnetic materials w/ a small air gap in between them



$$B_1 A_1 = B_2 A_2 = B_3 A_3$$

If $A_1 = A_2 = A_3 = A$, then $B_1 = B_2 = B_3 \rightarrow$ we can create a magnetic field inside an air gap

- * If only given $A_1 = A_3$, we can only assume $A_1 = A_2 = A_3 = A$ when the air gap is small enough to prevent any fringing magnetic fields (A_2 would be larger for large air gaps)

- Due to the principle of flux conservation, we can treat magnetically susceptible materials like wires in a magnetic circuit.

(We assume high μ_r materials effectively contain the magnetic flux). $R_{\text{tot}} = \frac{l}{\mu_0 \mu_r A}$

- This also assumes that fringing effects at sharp corners can be neglected.

Magnetic materials

- There is a limit to the magnetic flux density B we can generate using air-filled coils

$\rightarrow \uparrow I \rightarrow$ resistive heating $\rightarrow$ wires become hot $\rightarrow \uparrow R$ in coil

$\rightarrow$ exert forces (as $F = BIL$) $\rightarrow$ wires become in tension

$\rightarrow \uparrow n \rightarrow \downarrow d \rightarrow \downarrow A \rightarrow \uparrow R$ in coil

$\rightarrow \downarrow d \rightarrow$ mechanically weaker

$\rightarrow$ filling the coil filled w/ magnetic material can enhance the magnetic flux density B .

- The behaviour of materials when placed in a magnetic field is a consequence of three types of behaviour:

$\rightarrow$ paramagnetism: Internal magnetic fields line up and reinforce the external field
(Linear)

Atoms that have a net dipole moment. (Unpaired e^-)

$\rightarrow$ diamagnetism: Internal magnetic fields add up in ways that opposes the external field.
(Linear)

Atoms that have no net dipole moment. (No unpaired e^-)

$\rightarrow$ ferromagnetism: Internal fields are strongly interacting w/ each other lining up into magnetic domains, even in the absence of an external field.
(Non-linear)

- When subjected to an ext. magnetic flux density B_0 , there is an internally generated magnetic flux density I (intensity of magnetization) that add to the ext. field, given by

$$I = M_0 M$$

$$M = \frac{M_{\text{total}}}{V}$$

where M is the magnetization and M_{total} is the total magnetic moment of the material.

- The resultant magnetic flux density B is therefore given by,

$$B = B_0 + I$$

$$B = \mu_0 (H + M)$$

For Personal Use Only -bkwk2

Linear magnetic material. (Paramagnetic (diamagnetic)).

- For a linear magnetic material, the magnetization M is directly proportional to the magnetic field intensity of the ext. field H

$$M = \chi_m H$$

where χ_m is the magnetic susceptibility of the material.

- The resultant magnetic flux density B is therefore given by

$$\begin{aligned} B &= \mu_0 (H + M) \\ &= \mu_0 (H + \chi_m H) \\ &= \mu_0 H (1 + \chi_m) \end{aligned}$$

$$B = \mu_0 \mu_r H$$

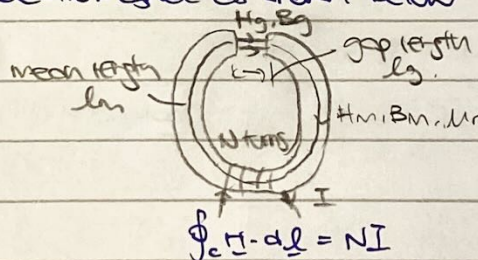
- The relative permeability μ_r and magnetic susceptibility χ_m are related by

$$\mu_r = 1 + \chi_m$$

- Paramagnetic and diamagnetic materials are soft (temporary) and weak ($\mu_r \approx 1$).
- Paramagnetic materials have a +ve magnetic susceptibility χ_m ; Diamagnetic materials have a -ve magnetic susceptibility χ_m .
 Curie's Law *material property*
- Paramagnetism is temperature dependent ($\chi_m = \frac{C}{T}$, where C is the Curie constant);
Diamagnetism is independent of temperature.

Magnetic flux in a linear electromagnet

- Consider a toroidal electromagnet as shown below



A/L

$$\oint_c H \cdot dl = NI$$

$$H_m l_m + H_g l_g = NI$$

Use B

$$\frac{B_m}{\mu_r \mu_0} l_m + \frac{B_g}{\mu_0} l_g = NI$$

cons. of flux

$$\Phi_m = \Phi_g$$

assume fringing effects are negligible.

$$B_m \Phi = B_g \Phi \rightarrow B_m = B_g$$

$$\therefore \frac{B_g}{\mu_r \mu_0} l_m + \frac{B_g}{\mu_0} l_g = NI$$

For large μ_r , $\frac{B_g}{\mu_r \mu_0} l_m \ll \frac{B_g}{\mu_0} l_g$ so $\frac{B_g}{\mu_r \mu_0} l_m + \frac{B_g}{\mu_0} l_g \approx \frac{B_g}{\mu_0} l_g$

$$\text{i.e. } B_g = \frac{NI \mu_r}{l_g}$$

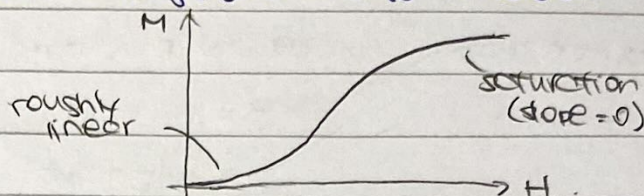
- This setup is used in tape recorder heads, where magnetization in magnetic tape (B) is related to the audio signal strength (I) $B_g \propto I$.

non-linear magnetic material (ferromagnetic). ^{above RTP} only Fe, Co, Ni

- For a non-linear magnetic material, the magnetization M is related to the magnetic field intensity of the ext. field H through a non-linear function f_n .

$$M = f_n(H)$$

- The function f_n follows the rough general shape as follows:



- Initially, M increases w/ H at an increasing rate, but it eventually flattens off as all of the magnetic domains available are aligned (saturation).

- the resultant magnetic flux density B is still given by

$$\begin{aligned} B &= \mu_0(H + M) \\ &= \mu_0 \mu_r H \end{aligned}$$

linear magnetic materials can also experience saturation at very large applied H but is practically always linear

but now that M varies non-linearly w/ H , μ_r is no longer constant.

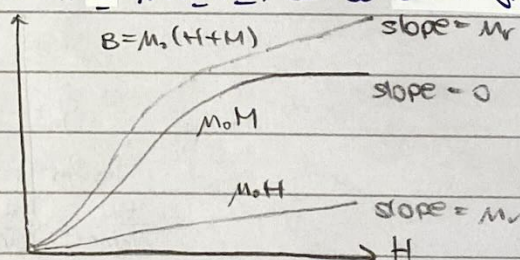
(we use differential susceptibility $\chi_m^d = \frac{dM}{dH}$ or apparent susceptibility).

- Ferromagnetic materials can be hard (permanent) and are strong ($\mu_r \times 10^4$).
- When heated above the Curie temperature T_c , ferromagnetic materials become paramagnetic (degree of alignment of atomic magnetic moments decreases). Above T_c , susceptibility χ_m is given by $\chi_m = \frac{C}{T - T_c}$ where C is the Curie constant

← Curie-Weiss Law

Relationship between B and M .

- The relationship between B and M . $B = \mu_0(H + M)$ can be shown graphically:



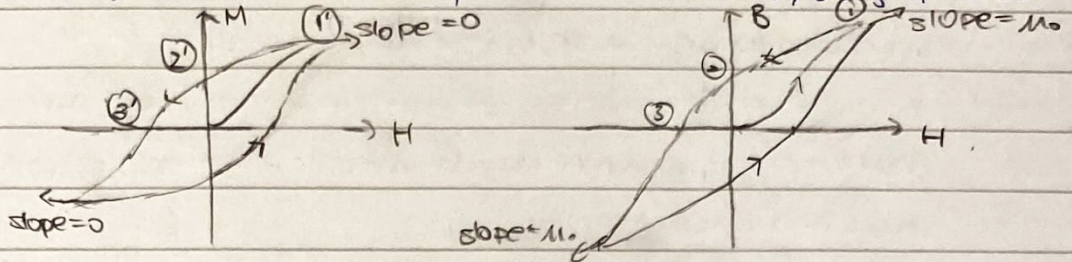
- At infinitely large H , B tends to infinity but M stays a constant value.

(so B does not saturate!)

- the $M-H$ curve is an intrinsic characteristic of the magnetic material.

Magnetic hysteresis

- Non-linear magnets may have memory (permanent magnetism), giving hysteresis.

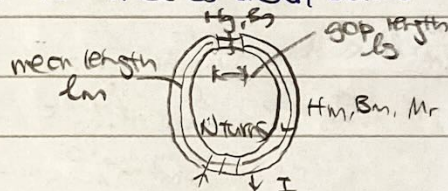


- The magnetic material is saturated at ①.
- At ②, H is zero but there is a remanent flux density B_r (y-intercept) *or retentivity/remnance*
- At ③, a -ve H is req. to reach zero B , this H is the coercivity H_c (x-intercept)
- The H at ② and ③ are different. They are the coercivity H_c and intrinsic coercivity H_{ci} respectively (for soft magnets, $H_c \approx H_{ci}$)

- + $M-H$ and $B-H$ curves are not necessarily symmetrical. It may be easier/harder to reach saturation depending on the crystallographic direction.
- The portion from ② to ③ is the demagnetization curve.

Magnetic flux in a non-linear electromagnet

- consider a toroidal electromagnet as shown below



A/L

$$\oint \vec{H} \cdot d\vec{l} = NI$$

$$H_m l_m + H_g l_g = NI$$

use B

$$H_m l_m + \frac{B_g}{\mu_0} l_g = NI$$

cons. of flux

$$\Phi_m = \Phi_g$$

assume fringing effects are negligible.

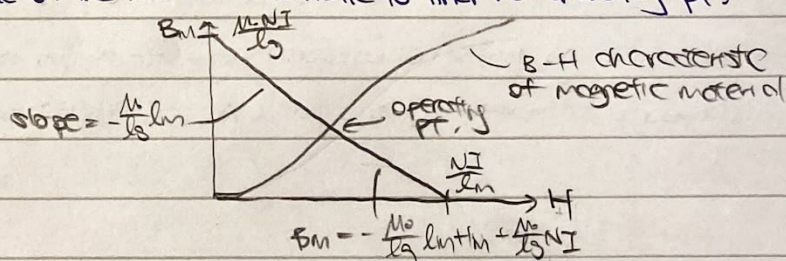
$$B_m A = B_g A \rightarrow B_m = B_g$$

$$\therefore H_m l_m + \frac{B_m}{\mu_0} l_g = NI$$

$$B_m = -\frac{\mu_0}{l_g} l_m H_m + \frac{\mu_0}{l_g} NI$$

load line

Plotting the load line on the $B-H$ characteristic to find the operating pt.



Magnetic flux in a permanent magnet

- A permanent magnet produces a magnetic flux even after any ext. magnetic field have been removed, w/ the max. remanent flux B_r .
- Permanent magnetic materials are expensive $\rightarrow$ rarely used alone $\rightarrow$ we usually combine a small piece of permanent magnetic material w/ soft iron pieces to channel the magnetic flux to the req. pt.
- By reducing the CSA of the air gap, the flux becomes more conc. (flux concn.).
- Consider a permanent magnet as shown below:



A/C

$$\oint H \cdot d\ell = NI$$

$$H_m l_m + H_i l_i + H_g l_g = 0 \quad \checkmark \quad NI = 0 \text{ for permanent magnets}$$

use B

$$H_m l_m + \frac{B_i}{\mu_0 \mu_i} l_i + \frac{B_g}{\mu_0} l_g = 0$$

Consv. of flux

$$\Phi_m = \Phi_i = \Phi_g \quad \checkmark \quad \text{assume fringing effects are negligible}$$

$$B_m A_m = B_i A_i = B_g A_g \rightarrow B_i = \frac{B_m A_m}{A_i} ; B_g = \frac{B_m A_m}{A_g}$$

$$\therefore H_m l_m + \frac{B_m A_m}{\mu_0 \mu_i A_i} l_i + \frac{B_m A_m}{\mu_0 A_g} l_g = 0$$

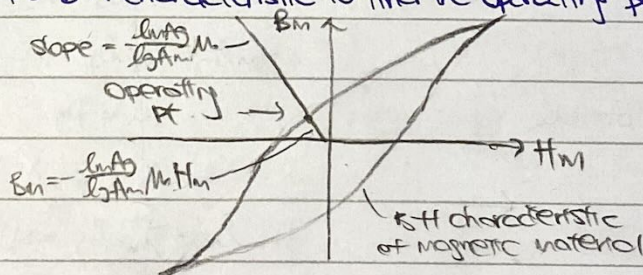
$$B_m A_m \left(\frac{l_i}{\mu_0 \mu_i A_i} + \frac{l_g}{\mu_0 A_g} \right) = -H_m l_m$$

for large μ_i , $\frac{l_i}{\mu_0 \mu_i A_i} \ll \frac{l_g}{\mu_0 A_g}$, so $\frac{l_i}{\mu_0 \mu_i A_i} + \frac{l_g}{\mu_0 A_g} \approx \frac{l_g}{\mu_0 A_g}$

$$\text{r.e.} \quad B_m A_m \frac{l_g}{\mu_0 A_g} = -H_m l_m$$

$$B_m = - \frac{l_m A_g}{l_g A_m \mu_0} H_m \quad \checkmark \quad \text{load line}$$

Plotting the load line on the B-H characteristic to find the operating pt.



* The permanent magnet operates on the demagnetisation section of the hysteresis curve

* We have assumed that B_i is prop. to H_i for soft iron. However, this is not strictly true for iron as μ_i is not constant. However as μ_i is very large $\sim 10^3$, the term has disappeared from the final answer (we can neglect any terms w/ $\mu_i > 1$).

Magnetostatic energy and forces

Energy stored in a magnetic field.

- Consider the work done δW in increasing the flux by $\delta \Phi$ against the induced emf V .

$$V = \frac{\delta \Phi}{\delta t}, \quad \delta \Phi = N A \delta B, \quad N I = H l. \quad P = IV \quad // \quad \delta W = P \delta t = I \frac{\delta \Phi}{\delta t} \delta t = I N A \delta B = H l A \delta B = V H \delta B$$

$$\therefore W = V \int_0^B H dB$$

work done per unit volume

$$\frac{W}{V} = \int_0^B H dB$$

area under the H-B curve.

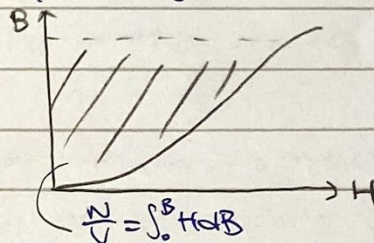
- For linear magnetic materials, B is prop. to H , $B = \mu_0 \mu_r H$, so

$$\frac{W}{V} = \int_0^B \frac{B}{\mu_0 \mu_r} dB$$

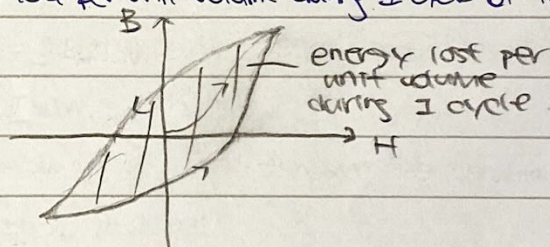
$$\frac{W}{V} = \frac{B^2}{2 \mu_0 \mu_r}$$

can be v. large as $\mu_0 \mu_r$ is v. small

- for non-linear magnetic materials, B and H are related by the B-H characteristic of the material, so we cannot simplify the integral analytically.



- If the material has hysteresis, the area inside the B-H loop is not accessible and this area represents the energy lost per unit volume during 1 cycle of the B-H loop.



force from an electromagnet

- Consider an electromagnet w/ a core of permeability $\mu_0 \mu_r$ wound w/ a coil of N turns carrying a current I . A "keeper" of the same material is placed across the poles.

- To calculate the force F needed to just remove the "keeper",

we equate the mechanical work done by the force F moving the "keeper" by a short distance δx to the magnetostatic energy stored in the new air gap.

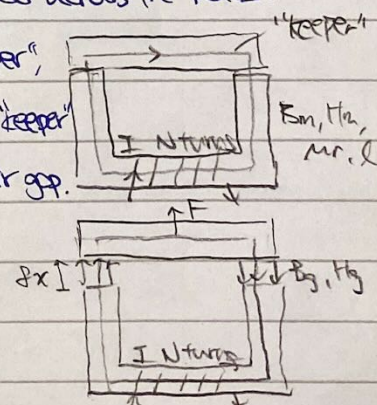
$$F \delta x = V \int_0^{B_g} H_g dB$$

$V = 2 A \delta x$ as A is the area of 1 gap, we have 2 gaps

$$F \delta x = 2 A \delta x \int_0^{B_g} \frac{B_g}{\mu_0} dB$$

$$F = 2 A \frac{B_g^2}{2 \mu_0}$$

$$\frac{F}{A} = \frac{B_g^2}{\mu_0}$$



cons. of flux $\Phi_m = \Phi_g \rightarrow B_m l = B_g A \rightarrow B_m = B_g \therefore \frac{F}{A} = \frac{B_m^2}{\mu_0}$

Assume same CSA A throughout

- * Assume δx is small so no fringing effects $\rightarrow \delta \Phi = 0 \rightarrow$ induced emf $V = 0 \rightarrow$ no need to account for the work done against the induced emf $\delta W = IV \delta t$.

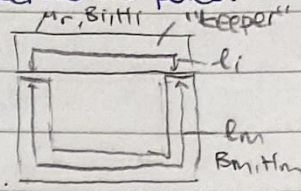
For Personal Use Only -bkwk2

Forces from a permanent magnet

- Consider a permanent magnet w/ a piece of soft iron (keeper) placed across the poles.

- To calculate the force F needed to just remove the "keeper", we

equote the mechanical work done by the force F in moving the "keeper" by a short distance δx to the magnetostatic energy stored in the new air gap.



$$F \delta x = \int_0^{B_g} H_g dB$$

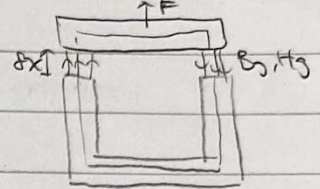
$V = 2A \delta x$ as
A is the area of 2

gaps, we have 2 gaps

$$F \delta x = 2A \delta x \int_0^{B_g} \frac{B_g}{\mu_0} dB$$

$$F = 2A \frac{B_g^2}{2\mu_0}$$

$$\frac{F}{A} = \frac{B_g^2}{\mu_0}$$



Assume constant CSA
A throughout

consu. of flux $\phi_m = \phi_g \rightarrow B_m A = B_g A \rightarrow B_m = B_g \therefore \frac{F}{A} = \frac{B_m^2}{\mu_0}$

* Assume δx is small so no fringing effects $\rightarrow d\phi = 0 \rightarrow$ induced emf $V = 0 \rightarrow$ no need to account for the work done against the induced emf $dW = IV dt$.

Forces from electromagnets vs permanent magnets.

- The expression for the magnetostatic force (per unit area) is the same for both electromagnets

and permanent magnets: $\frac{F}{A} = \frac{B_m^2}{\mu_0}$

- The only difference is how we obtain an expression/value for B_m .

using techniques mentioned before.

$\rightarrow$ Linear electromagnet: Use $\oint_C H_m \cdot dl = NI$ to find H_m

Use $B_m = \mu_r \mu_0 H_m$ to find B_m

$\rightarrow$ Non-linear electromagnet: Use $\oint_C H_m \cdot dl = NI$ to find H_m

Use B-H characteristic to find B_m

$\rightarrow$ Permanent magnet: Use $\oint_C H_m \cdot dl = NI$ to find an expression w/ H_m, H_i

Use $B_i = \mu_0 \mu_i H_i$ to express in terms of B_i

Use consu. of flux to express in terms of B_m, H_m only $\rightarrow$ load line

Use load line + B-H characteristic to find the operating pt. (B_m)